

Phase 2 Final Audit — DEPLOYMENT READY Report

Smart Lead Qualification — Vyapar AI Employee Platform

Audit type: Read-only final audit (no code changes)

Date: June 1, 2026

Verdict: DEPLOYMENT READY Phase 1 + Phase 2

Executive summary: Phase 2 is fully implemented on disk. All required files exist, wiring is correct, imports resolve under static analysis, and the personal memory path is unchanged and still runs before the lead pipeline. Deploy Phase 1 and Phase 2 files atomically.

1. Audit Checklist

#	Check	Result
1	All Phase 2 files exist	PASS
2	No missing imports (static analysis)	PASS
3	No duplicate active company configuration	PASS
4	lead_manager.py connected to ai_employee_engine.py	PASS
5	lead_extractor.py connected	PASS
6	admin_notifier.py connected	PASS
7	Memory system still works	PASS
8	DEPLOYMENT_READY report generated	PASS

2. Phase 2 File Inventory

New files

File	Purpose	Status
lead_extractor.py	Buying intent, signals, scoring, stage derivation	Present
lead_manager.py	upsert_lead, sales memory, export	Present
admin_notifier.py	Log-only admin alert placeholder	Present
admin_export_leads.py	CLI lead export to exports/	Present
tests/test_lead_phase2.py	7 unit tests for lead rules	Present

Modified files

File	Change	Status
memory_db.py	Lead model (~25 columns)	Present
products.py	find_best_product_match, format_alternative_product, include_full_catalog	Present
prompts.py	Sales hints, lead_block, sales_memory_block	Present
intent_engine.py	buying_intent, coverage_inquiry, lead_stage	Present
ai_employee_engine.py	Full lead pipeline integration	Present
env.example	LEAD_ACTIVE_DAYS, ADMIN_ALERT_*	Present
README_DEPLOY.md	Phase 2 deploy and test docs	Present

Phase 1 dependencies (required at deploy)

File	Status
company_profiles.json, company_manager.py, business_settings.py	Present
main.py, memory.py, memory_extractor.py, openai_engine.py	Present

3. Import Integrity

All cross-module imports resolve to existing symbols. No new requirements.txt dependencies for Phase 2.

Module	Imports from	Status
lead_extractor.py	memory_extractor (city, name, phone, etc.)	OK
lead_manager.py	memory_db.Lead, get_session	OK
ai_employee_engine.py	admin_notifier, lead_extractor, lead_manager, products, prompts, memory	OK
admin_export_leads.py	company_manager, lead_manager.export_leads	OK
tests/test_lead_phase2.py	lead_extractor (7 symbols)	OK

NOTE Runtime import test not executed on audit machine (Python not installed). Run on deploy target:

```
python -m unittest tests.test_lead_phase2 -v
python -c "import ai_employee_engine; print('OK')"
```

4. Company Configuration

File	Role	Referenced in code?
company_profiles.json	Single source of truth (Phase 1)	Yes — company_manager.py only
company_profile.json	Legacy flat HONS schema	No — zero .py references

business_settings.py and products.py load via company_manager. No hardcoded HONS strings in Python.

Non-blocking: Delete company_profile.json when convenient to avoid operator confusion.

5. Integration Wiring

5.1 lead_manager ↔ ai_employee_engine

Function	Usage
get_active_lead	Load existing lead before extraction
upsert_lead	Persist when should_process_lead(bundle)
lead_to_prompt	Inject lead context into system prompt
sales_memory_to_prompt	Inject sales conversation memory
update_sales_memory	After sales-mode reply
export_leads	Via admin_export_leads.py CLI

5.2 lead_extractor ↔ ai_employee_engine

Pipeline order in generate_employee_reply:

1. Memory extraction (unchanged)
2. Direct memory short-circuit (unchanged)
3. get_active_lead → extract_lead_bundle → conditional upsert_lead
4. find_best_product_match + sales mode logic
5. Prompt composition with lead blocks → OpenAI → admin notify

Qualified rule (approved):

qualified requires (**phone OR WhatsApp**) AND (**location OR speed**).

Score alone cannot create qualified — max stage without contact is interested.

5.3 admin_notifier ↔ ai_employee_engine

maybe_notify_admin(lead=lead, company_id=tenant_id) called after reply generation. Log-only when ADMIN_ALERT_ENABLED=true; default false.

6. Memory System — Preserved

Step	Function	Status
Save user turn	save_chat_turn	Unchanged
Read + extract facts	read_memory, extract_memory_facts	Unchanged
Persist facts + context	update_memory_from_facts, save_context	Unchanged
Direct self-query answer	_direct_memory_answer	Bypasses lead pipeline
Memory in prompt	memory_to_prompt	Unchanged

Phase 2 adds leads table only. init_db() on startup creates it automatically.

7. Architecture (Post Phase 2)

```
Telegram → main.py → ai_employee_engine.py
├─ memory_extractor + memory.py (unchanged)
├─ lead_extractor (NEW)
│   └─ compute score + derive stage
├─ lead_manager (NEW) → leads table
├─ company_manager ← company_profiles.json
├─ products.find_best_product_match (NEW)
├─ sales_mode? → single alternative, no catalog dump
├─ prompts: lead_block + sales_memory_block
├─ openai_engine (unchanged)
└─ admin_notifier placeholder (NEW)
```

8. Deployment Checklist

Required environment variables (Render dashboard)

```
COMPANY_ID=hons
LEAD_ACTIVE_DAYS=30
ADMIN_ALERT_ENABLED=false
ADMIN_ALERT_MIN_SCORE=80
ADMIN_ALERT_STAGES=hot,qualified
```

render.yaml does not declare Phase 2 vars; code defaults apply. Set explicitly in Render for clarity.

CRITICAL: Atomic deploy

Deploy **all Phase 1 + Phase 2 files in one deploy**. Partial deploy (e.g. new ai_employee_engine.py with old products.py missing company_id=) caused production TypeError previously.

Post-deploy smoke tests

Test message	Expected
Internet jodnu cha	Lead signals, sales mode
Baluwatar ma 500 mbps internet jodnu cha	Stage interested, not qualified
Mero phone 9801234567 ho, Baluwatar ma 100 mbps lina cha	Stage qualified
Support hours kati ho?	No lead processing
Mero naam ke ho?	Direct memory answer

Logs to watch: LEAD_SIGNALS_DETECTED, LEAD_UPSERTED, EMPLOYEE_REPLY_GENERATED

Export: python admin_export_leads.py → exports/leads_export_{company_id}_{timestamp}.json

9. Issues & Recommendations

Item	Severity	Action
company_profile.json duplicate	Low	Delete when ready (not wired)
render.yaml missing Phase 2 env	Low	Add vars in Render dashboard
Unit tests not run on audit machine	Medium	Run unittest before/after deploy
No git repo on disk	Info	Use full-file deploy or init git

Final verdict: DEPLOYMENT READY

Deploy atomically with Phase 1, set env vars, restart service (creates leads table), run unit tests and Telegram smoke tests.